BPB/MBE Paper III: CMB dictionary selection and Weyl–lensing closure

v0.4.1 HAL corpus-aware source release

Fabien Pelletier
BPB/MBE corpus release

May 2026

Abstract

This paper tests the BPB/MBE CMB dictionary against three CMB-facing diagnostics: ACT DR6 lensing, Planck-lite high-ℓ TTTEEE, and Planck CMB-marginalized lensing. The result is not presented as an official full Planck nuisance/foreground likelihood or an official combined ACT+Planck likelihood. It is a dictionary-selection paper. The naive vanilla late-density CMB mapping fails catastrophically, while the strict face-minus branch remains viable at diagnostic level and prefers a modest positive Weyl-auto/lensing amplitude. The Planck-lite data/prior decomposition is central to this interpretation: the strict branch improves the acoustic-shape data term while the remaining penalty is dominated by the calibration-normalization prior. In the locked source tables, the ACT lenslike diagnostic gives $\chi^2_{\mathrm{ACT}} = 227.520768$ for the vanilla mapping, versus 17.470021 for the strict Kprimary unit-amplitude branch and 14.153599 after scalar Weyl-auto profiling. The Planck-lite high-ℓ proxy gives $\chi^2 = 749.191768$ for the strict C_{s_surv0} branch, better than the Planck sanity point in that lite diagnostic, while the vanilla mapping gives $\chi^2 = 8851.940269$. The joint ACT+Planck scalar lensing diagnostic closes with one common positive amplitude $\mathrm{amp}_\phi = 1.057$, $\Sigma_{\mathrm{eff}} = 1.028105$, and a joint penalty of only $\Delta\chi^2 = 0.825232$ relative to separate scalar minima. The guardrail is explicit: Paper III selects the surviving CMB dictionary and Weyl/lensing hierarchy, but does not claim final CMB Bayesian model selection.

1 Corpus placement and release navigation

Paper III is the CMB/Weyl descendant of Paper 0 in the BPB/MBE corpus. Paper 0 defines the dictionary and survival factor; Paper I tests low-redshift background/growth/lensing; Paper II tests galactic/SPARC descendants; Paper III asks whether the same dictionary can pass CMB-facing gates without collapsing primary CMB, matter growth, and Weyl lensing into a single amplitude.

The release is navigated by the root artefacts `00_README_CORPUS`, `00_MANIFEST_CORPUS.csv`, `00_CLAIMS_INDEX.csv`, and `00_REVIEWER_ROUTE`. This source package includes the Paper-III source-lock tables under `source_locks/` and canonical table/figure inventories under `tables/`.

2 Question tested

The question is not whether a final, full official Planck MCMC has been completed. It has not. The question tested here is sharper: does the BPB/MBE CMB dictionary fail already at diagnostic level, or does the strict face-minus branch survive while the vanilla mapping fails?

The locked answer is branch-selective. The vanilla late-density mapping is the failed CMB dictionary. The strict face-minus dictionary is the surviving branch across ACT lensing, Planck-lite high-ℓ TTTEEE, native lowT/lowE support, and Planck CMB-marginalized lensing diagnostics. A key point, developed in Section 7, is that the residual strict-branch Planck-lite penalty is not an acoustic-shape failure: it is primarily an A_{Planck} calibration-normalization prior penalty, while the high-ℓ data term itself improves relative to the Planck sanity point.

3 Dictionary branches and guardrails

We distinguish three response objects throughout the paper:

1. the primary-CMB dictionary, which controls high-ℓ acoustic structure;
2. the low-redshift matter response, already tested in Paper I;
3. the Weyl/lensing response, tested here through ACT and Planck lensing windows.

These are not interchangeable amplitudes. The strict face-minus branch is allowed to require a modest positive Weyl-auto/lensing amplitude while preserving the high-ℓ acoustic dictionary.

The following phrases are forbidden for this version: "official full Planck likelihood", "official combined ACT+Planck likelihood", and "final CMB Bayesian model selection". The correct phrasing is "Planck-lite/proxy high-ℓ diagnostic", "Planck CMB-marginalized lensing official-formula diagnostic", and "joint scalar-amplitude diagnostic closure".

4 Audit summary and source-lock pointer

Paper III is backed by a source-lock gate chain, but the internal gate inventory is kept out of the main argument. The body uses only the scientific outputs: ACT lensing, Planck-lite high-ℓ TTTEEE, the Planck-lite data/prior decomposition, native lowT/lowE support, Planck CMB-marginalized lensing, and the joint scalar Weyl-auto closure. The detailed P3-A0–P3-A4b computational inventory is recorded in Appendix A and in the packaged `source_locks/` CSV files.

5 ACT DR6 lensing gate

The ACT DR6 lenslike diagnostic is the Weyl-sector stress gate for the CMB dictionary. It does not by itself decide the full corpus. It tests whether the projected CMB dictionary yields a viable lensing/Weyl spectrum.

Table 1: ACT DR6 lenslike diagnostic. The vanilla late-density mapping is the failure control. The strict face-minus/Kprimary branch remains close to the Planck sanity point and improves under scalar Weyl-auto profiling.

Case	$\mathrm{amp}_{\Phi\Phi}$	Σ_{eff}	χ^2_{ACT}
Planck PR4 sanity	1.000	1.000000	13.951140
Vanilla late-density	1.000	1.000000	227.520768
Strict Kprimary	1.000	1.000000	17.470021
Strict Kprimary, CMB8 amp	1.042	1.020858	14.156104
Strict Kprimary scan best	1.043	1.021274	14.153599

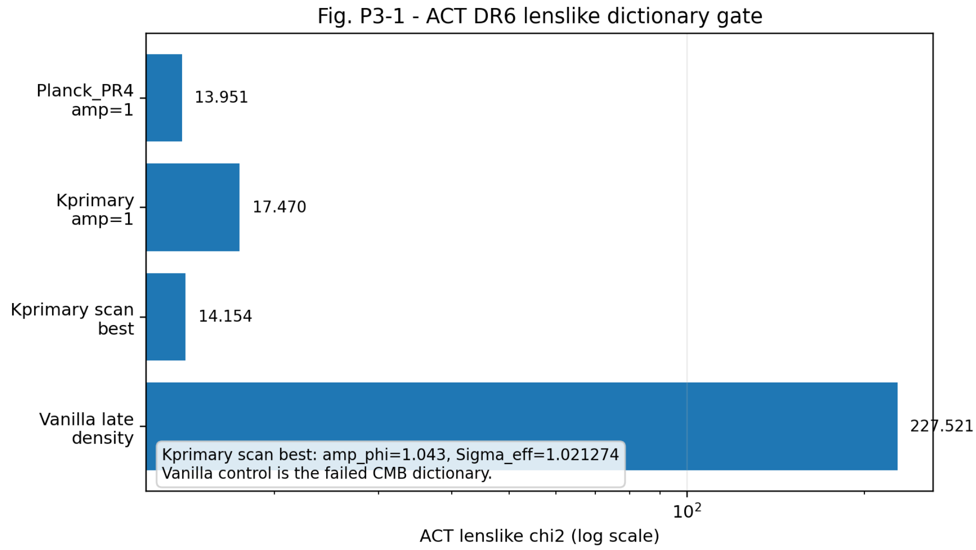

Figure 1: ACT DR6 lenslike CMB dictionary gate. The vanilla late-density CMB mapping is catastrophically disfavoured, while the strict face-minus/Kprimary dictionary remains close to the Planck sanity point and improves under a small positive Weyl-auto amplitude profile.

6 Planck-lite high-ℓ TTTEEE gate

The Planck-lite high-ℓ diagnostic tests whether the branch preserves the high-ℓ acoustic shape under a lite/proxy treatment. It is not a full official Planck foreground/nuisance likelihood.

Table 2: Planck-lite high-ℓ TTTEEE proxy. The strict face-minus branch does not show a shape crash; the vanilla mapping does.

Case	n_s	As scale	Best χ^2	$\Delta\chi^2$ vs Planck
Planck PR4 sanity	0.972	0.986	765.869165	0.000000
Strict C_{s_surv0}	0.982	0.976	749.191768	-16.677397
$C_{Z\mathrm{best}}$	–	–	777.286828	11.417663
Vanilla control	–	–	8851.940269	8086.071104

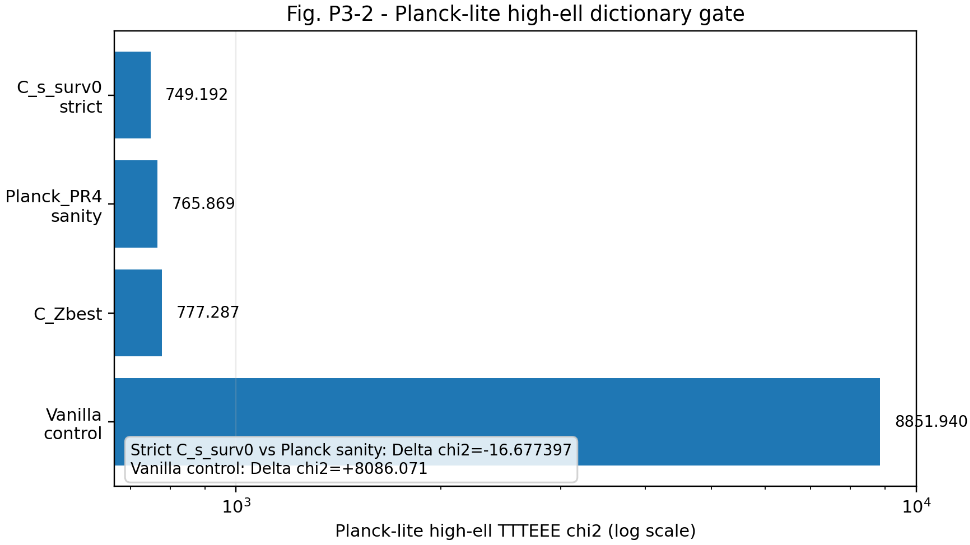

Figure 2: Planck-lite high-ℓ TTTEEE proxy. The strict C_{s_surv0} dictionary improves the profiled high-ℓ χ^2 relative to the Planck sanity point in this lite diagnostic, while the vanilla dictionary fails by thousands in χ^2.

7 Planck-lite data/prior decomposition

P3-A4b locks the interpretation of the Planck-lite residual penalty. In the fine n_s rows, the strict branch improves the TT/TE/EE data term but is penalized by the A_{Planck} calibration-normalization prior.

Table 3: Planck-lite fine n_s data/prior decomposition. The strict branch is shape-favourable in the data term and prior-penalized in calibration normalization.

Case	n_s	A_{Planck}	χ^2_{data}	χ^2_{total}
Planck sanity best	0.970	0.988996	771.942231	791.314859
Strict C_{s_surv0} best	0.976	0.982163	766.888198	817.791438
Difference strict - Planck	–	–	-5.054033	26.476578
Prior contribution difference	–	–	–	31.530611

This supports the wording "better acoustic-shape/data term, calibration-normalization penalty", not "full Planck closure".

8 Low-ℓ and τ support

The high-ℓ result is supported by the native lowT/lowE plus τ/A_s grid. The source-lock table gives Planck total $\chi^2 = 1182.553918$, strict C_{s_surv0} total $\chi^2 = 1164.645118$, and $\Delta\chi^2_{\mathrm{total}} = -17.908799$, with $\Delta\chi^2_{\mathrm{high}\ell} = -17.386802$ and $\Delta\chi^2_{\mathrm{lowT+lowE}} = -0.521997$. Thus the supporting low-ℓ modules do not generate a compensating failure.

9 Planck CMB-marginalized lensing diagnostic

The Planck lensing gate uses the CMB-marginalized lensing official-formula diagnostic at scalar-amplitude level. The strict branch prefers a modest positive lensing/Weyl-auto amplitude.

Table 4: Planck CMB-marginalized lensing diagnostic with scalar amplitude profiling.

Case	Best amp_ϕ	Σ_{eff}	Best χ^2
Planck PR4 sanity	1.020	1.009950	9.181939
Strict C_{s_surv0}	1.075	1.036822	8.742460
Vanilla control	0.840	0.916515	20.544514

The scalar-amplitude result is enough for this paper's claim: Planck lensing does not kill the strict branch, while vanilla remains poor. The supporting three-band kernel forensic is recorded in source locks but is not promoted to a final physical kernel.

10 Joint ACT+Planck scalar Weyl-auto closure

The common-amplitude test asks whether ACT and Planck lensing can be served by one scalar Weyl-auto response. The answer is yes at diagnostic level: the joint best is close to the separate minima and has only a small penalty.

Table 5: Joint ACT+Planck scalar lensing closure. This is a scalar-amplitude diagnostic, not an official combined likelihood.

Selector	amp_ϕ	Σ_{eff}	χ^2_{ACT}	$\chi^2_{\mathrm{Planck\ lens}}$
ACT-only best	1.043	1.021274	14.153599	10.198317
Planck-only best	1.075	1.036822	15.908702	8.742460
Joint scalar best	1.057	1.028105	14.478027	9.243264

The separate-minima sum is 22.896060. The joint scalar best has $\chi^2_{\mathrm{joint}} = 23.721292$, so the joint penalty is $\Delta\chi^2 = 0.825232$.

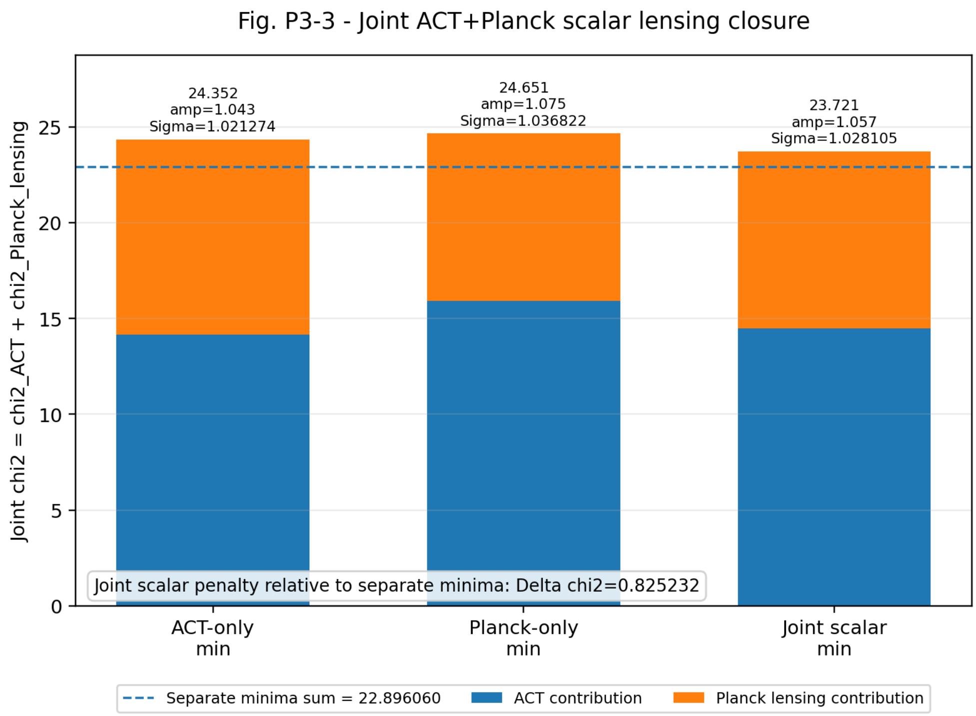

Figure 3: Scalar-amplitude closure for ACT DR6 and Planck CMB-marginalized lensing. ACT-only and Planck-only windows prefer nearby positive amplitudes, and one common $\mathrm{amp}_\phi = 1.057$ gives a joint penalty of only $\Delta\chi^2 = 0.825232$ relative to separate minima.

11 Synthesis

The CMB result is sharper than a generic consistency statement. The vanilla late-density dictionary is the failed branch. The strict face-minus dictionary is the surviving branch across the audited diagnostics. Because the lensing closure requires a modest positive Weyl-auto amplitude, and because ACT, Planck-lite and Planck lensing probe different windows, Paper III states the response hierarchy explicitly: the primary-CMB dictionary, the low-redshift matter response and the Weyl/lensing response are not interchangeable amplitudes.

12 Claim boundaries

Allowed claims:

- ACT DR6 lenslike kills the vanilla CMB dictionary but not the strict face-minus branch.
- Planck-lite high-ℓ TTTEEE shows no acoustic-shape crash for strict face-minus; vanilla fails catastrophically.
- The Planck-lite strict-branch residual penalty is dominated by A_{Planck} calibration-normalization prior, while the data term is favourable.
- Planck CMB-marginalized lensing and ACT lensing both prefer a modest positive Weyl-auto amplitude.
- A common scalar amplitude closes ACT+Planck lensing at diagnostic level with small joint penalty.

Forbidden claims:

- full official Planck nuisance/foreground MCMC closure;
- official combined ACT+Planck likelihood closure;
- final CMB Bayesian model selection;
- final derived Weyl-window kernel.

A Source-lock gate inventory

The computational gate chain is:

- P3-A0: status audit of CMB artefacts without touching neutron-star outputs;
- P3-A1/P3-A1b: source match and joint lensing extraction patch;
- P3-A2: canonical table and figure inventory;
- P3-A3: wording and guardrail lock;
- P3-A4: canonical figure generation;
- P3-A4b: Planck-lite data/prior decomposition lock.

All gates are source-locked in the packaged CSV and markdown artefacts. This appendix is an audit route, not an additional physics claim.

B Data and reproducibility inventory

The source package contains:

- `source_locks/paperIII_cmb_claim_source_lock.csv`;
- `source_locks/paperIII_cmb_claim_source_lock_p3a1b_jointlens_patch.csv`;
- `source_locks/paperIII_cmb_claim_source_lock_p3a4b_plancklite02.csv`;
- `tables/paperIII_cmb_p3a2_canonical_tables.csv`;
- `tables/paperIII_cmb_p3a3_claim_wording.csv`;
- `figures/fig_P3_1_ACT_lensing_gate.png`;
- `figures/fig_P3_2_PlanckLite_high_ell.png`;
- `figures/fig_P3_3_joint_lensing_scalar_closure.png`.

C Version notes

This v0.4.1 source release is corpus-aware. It applies a text-only hotfix to foreground the Planck-lite data/prior decomposition, move the detailed source-lock gate chain into the appendix, and make the ACT DR6 lensing references explicit. The v0.4 release was corpus-aware. It integrates the P3-A0–P3-A4b source locks, adds explicit corpus navigation, and keeps the CMB guardrails in the body rather than only in the release index. No claim of full official Planck likelihood or official ACT+Planck combined likelihood is made.

References

[1] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, A&A 641, A6 (2020).

[2] F. J. Qu et al., *The Atacama Cosmology Telescope: a measurement of the DR6 CMB lensing power spectrum and its implications for structure growth*, Astrophys. J. 962, 112 (2024).

[3] M. S. Madhavacheril et al., *The Atacama Cosmology Telescope: DR6 gravitational lensing map and cosmological parameters*, Astrophys. J. 962, 113 (2024).

[4] A. Lewis, A. Challinor and A. Lasenby, *Efficient computation of CMB anisotropies in closed FRW models*, Astrophys. J. 538, 473 (2000).

[5] Planck Collaboration, *Planck likelihood and CMB-marginalized lensing products*, Planck legacy likelihood documentation.